

SINOP, Turkey

WMO: 170260

Lat: 42.0297N

Lon: 35.1543E

Elev: 32

StdP: 100.94

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.4	1.6	-5.5	2.4	4.4	-3.7	2.8	4.6	11.6	3.3	10.6	3.8	4.1	10	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.2	28.9	23.8	28.0	23.5	27.2	23.0	25.6	27.6	24.8	26.9	24.1	26.3	3.4	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	20.2	27.0	24.1	19.1	26.4	23.4	18.2	25.8	79.1	27.9	75.7	27.0	72.9	26.4	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	7.8	6.5	-1.3	30.7	1.8	0.8	-2.6	31.2	-3.7	31.7	-4.7	32.1	-6.0	32.7		
(5)			-2.3	26.1	2.0	1.1	-3.7	26.9	-4.9	27.6	-6.0	28.2	-7.4	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	7.7	7.2	8.4	11.4	16.1	20.8	23.7	24.5	20.9	17.0	12.7	9.5		
	DBStd	6.75	3.41	3.35	3.30	3.24	2.93	2.10	1.60	1.71	2.41	3.20	3.32	3.49		
	HDD10.0	333	89	90	71	18	0	0	0	0	0	1	13	52		
	HDD18.3	1760	331	312	307	210	81	4	0	0	7	65	171	273		
	CDD10.0	2172	16	12	23	59	191	324	426	449	327	216	93	37		
	CDD18.3	558	0	0	0	2	13	78	167	191	84	22	2	0		
	CDH23.3	2479	0	0	1	6	24	186	849	1179	192	40	3	0		
	CDH26.7	289	0	0	0	1	2	9	84	176	10	7	0	0		
(14) Wind	WSAvg	2.7	2.9	3.0	3.2	2.9	2.4	2.5	2.8	2.6	2.4	2.4	2.6	2.8		
(15) Precipitation	PrecAvg	650	74	45	41	39	37	29	31	47	58	87	80	84		
	PrecMax	990	148	128	93	80	115	83	182	225	126	324	243	176		
	PrecMin	333	2	5	0	7	7	5	0	0	3	17	12	24		
	PrecStd	152	33	25	23	20	23	18	37	49	33	63	49	42		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.2	18.2	21.2	24.2	25.7	27.9	29.7	30.6	28.0	27.1	23.3	20.2		
		MCWB	10.6	10.8	12.1	14.5	19.2	21.4	23.7	25.3	22.7	17.7	13.8	12.1		
	2%	DB	16.0	15.5	17.1	20.4	23.6	26.5	28.5	29.3	26.3	24.1	20.2	17.6		
		MCWB	9.7	9.6	10.8	13.9	18.1	21.6	23.2	24.7	21.5	18.2	13.9	11.5		
	5%	DB	14.3	13.7	14.9	18.0	21.8	25.4	27.6	28.4	25.3	22.5	18.4	15.9		
		MCWB	9.1	9.0	10.0	13.0	17.7	20.8	23.1	24.2	21.1	18.5	14.2	10.8		
	10%	DB	12.5	11.8	13.0	15.8	20.4	24.3	26.8	27.6	24.4	21.1	16.9	14.4		
		MCWB	8.5	8.3	9.5	11.9	17.0	20.5	23.0	23.7	20.6	18.2	13.6	10.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.9	13.3	17.0	22.3	24.6	26.4	27.0	24.2	20.9	17.6	13.7		
		MCDB	15.5	15.4	18.0	19.6	22.9	25.7	27.7	28.6	26.7	22.9	18.9	16.9		
	2%	WB	10.6	10.7	11.8	14.8	19.8	23.2	25.2	26.0	23.1	20.2	16.2	12.6		
		MCDB	14.3	13.9	15.2	19.2	21.4	24.9	27.1	28.2	25.0	22.1	17.4	15.4		
	5%	WB	9.8	9.7	10.7	13.4	18.6	22.2	24.4	25.2	22.2	19.4	15.1	11.8		
		MCDB	12.7	12.5	13.7	17.1	20.5	24.1	26.5	27.4	24.2	21.3	16.8	14.3		
	10%	WB	9.1	8.8	9.8	12.3	17.5	21.4	23.8	24.6	21.4	18.4	14.1	10.9		
		MCDB	11.3	10.8	12.7	15.0	19.7	23.4	25.9	26.8	23.5	20.5	16.2	13.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.9	4.4	4.7	4.9	5.2	5.4	5.2	5.2	5.0	4.6	4.5	4.2		
		MCDBR	6.8	8.1	8.6	8.7	7.4	6.9	6.1	6.1	6.0	6.4	6.9	6.3		
	5% WB	MCWBR	3.4	4.1	4.2	4.4	4.5	4.1	3.5	3.7	3.5	3.4	3.7	3.5		
		MCWBR	5.5	6.7	7.4	7.8	6.8	5.9	5.4	5.6	5.4	5.3	5.3	5.1		
(40) Clear-Sky Solar Irradiance	taub	taub	0.352	0.384	0.422	0.461	0.434	0.413	0.426	0.435	0.410	0.397	0.362	0.349		
		taud	2.364	2.265	2.174	2.100	2.265	2.375	2.356	2.319	2.375	2.357	2.398	2.383		
	Ebn at Noon	Ebn at Noon	777	804	815	811	845	863	846	826	822	783	759	745		
		Edh at Noon	84	108	133	153	133	120	121	122	108	98	81	77		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.51	2.25	3.46	4.87	5.91	6.88	6.96	6.05	4.52	2.93	1.95	1.35		
	RadStd	RadStd	0.14	0.28	0.25	0.49	0.56	0.44	0.34	0.32	0.34	0.27	0.21	0.16		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.50	N/A	N/A	N/A	N/A	N/A	-82	-143	+106	+66

Nomenclature: See separate page